



Adding a Twist

Shifted twisted Yangians and affine Grassmannian islices

Kang Lu

joint work with Weiqiang Wang and Alex Weekes (arXiv:2510.10652, arXiv:2512.19998)



June 18, 2026



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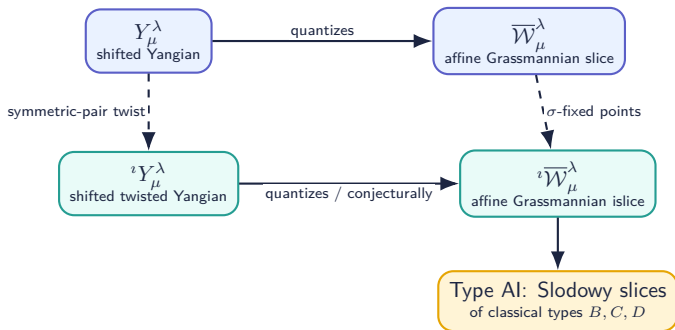
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The talk in one picture



Guiding slogan

Ordinary shifted Yangians see affine Grassmannian slices [KWWY,BFN]; shifted twisted Yangians see the corresponding symmetric-pair fixed loci [Li,LWW,Nakajima].

Part I: The ordinary story

- ▶ Loop groups and affine Grassmannian slices
- ▶ Shifted Yangians and GKLO maps
- ▶ Type A dictionary with Slodowy slices

Part II: Add the twist

- ▶ Quasi-split symmetric pairs
- ▶ Fixed point loci: islices
- ▶ Shifted twisted Yangians and quantization

Endgame

In type A1, affine Grassmannian islices match suitable nilpotent Slodowy slices of types B, C, D ; the expected Coulomb-branch model is **ortho-symplectic** or **hybrid**.

The ordinary Yangian–slice dictionary

Adding a twist to the Yangian–slice dictionary

Affine Grassmannian slices: the geometric side

Let G be the adjoint group of a simply-laced Lie algebra $\mathfrak{g}$. For a coweight μ , write $z^\mu \in T((z^{-1}))$ and set

$$\mathcal{W}_\mu = U_1^+[[z^{-1}]] T_1[[z^{-1}]] z^\mu U_1^-[[z^{-1}]] \subset G((z^{-1})).$$

e.g. $\mathcal{W}_0 = G[[z^{-1}]]$

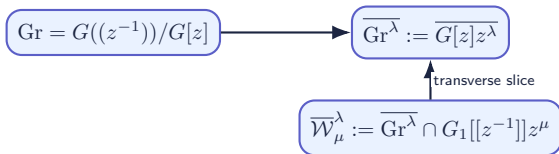
Generalized affine Grassmannian slices

For coweights $\lambda \geq \mu$, with λ dominant,

$$\dim \overline{\mathcal{W}}_\mu^\lambda = 2\langle \lambda - \mu, \rho \rangle$$

$$\overline{\mathcal{W}}_\mu^\lambda := \mathcal{W}_\mu \cap \overline{G[z]z^\lambda G[z]} \subset \mathcal{W}_\mu.$$

These are Poisson varieties and play the role of transverse slices in the affine Grassmannian.



Shifted Yangians: the algebraic side

The Yangian $Y := Y(\mathfrak{g})$ is a filtered deformation of $U(\mathfrak{g}[z])$. For a dominant coweight μ , one has a shifted Yangian Y_μ .

Generators

Drinfeld-type generators:

$$E_i^{(r)}, \quad F_i^{(r)}, \quad H_i^{(s)}$$

with shifts in the allowed values of s determined by μ .

Two filtrations

Different filtrations make different classical limits visible:

$$\mathrm{gr} Y_\mu \rightsquigarrow \mathbb{C}[\mathcal{W}_\mu].$$

Classical limit

The shifted Yangian is the quantization of the affine Grassmannian slice before imposing any involution.

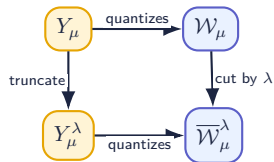
The ordinary theorem package

KWWY / BFN picture

For dominant μ , the shifted Yangian Y_μ quantizes $\mathcal{W}_\mu$:

$$\text{gr } Y_\mu \cong \mathbb{C}[\mathcal{W}_\mu]$$

as graded Poisson algebras.



GKLO philosophy

Difference-operator representations produce quotients Y_μ^λ that see the finite-dimensional slices $\overline{\mathcal{W}}_\mu^\lambda$.

Type A dictionary

When $G = PGL_N$, the slice picture admits a classical incarnation.

Nilpotent geometry		Affine Grassmannian geometry
$\overline{\mathcal{O}_\lambda}$ $\cap$ $\mathcal{N}_{\mathfrak{sl}_N}$	$\simeq$ $\simeq$ [Lusztig]	$\overline{\mathcal{W}}_0^\lambda$ $\cap$ $\overline{\mathcal{W}}_0^{N\varpi_1^\vee}$ $\overline{\mathcal{W}}_\mu^{N\varpi_1^\vee}$ $\overline{\mathcal{W}}_\mu^\lambda$
$e + \ker ad f =:$ e : Jordan form μ $\{e, h, f\}$ $\mathfrak{sl}_2$ -triple	Slodowy slice S_μ $\overline{\mathcal{O}_\lambda} \cap S_\mu$ $\simeq$ $\simeq$ [Mirkovic-Vybornov]	strata

Why this matters

The affine Grassmannian slice is not an abstract object only: in type A, it is the familiar transverse Slodowy-slice geometry in disguise.

Adding the symmetric-pair twist

Adding a twist to the Yangian–slice dictionary

Symmetric pairs and Satake diagrams

Let ω be the Chevalley involution and let τ be a diagram automorphism with $\tau^2 = 1$. The pair

$$\begin{aligned} e_i &\leftrightarrow -f_i \\ h_i &\rightarrow -h_i \end{aligned}$$

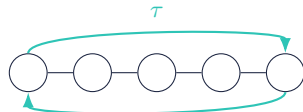
$$\theta = \omega \circ \tau, \quad \mathfrak{g}^\theta \subset \mathfrak{g}$$

encodes a quasi-split symmetric pair.

Satake input

A quasi-split Satake diagram records:

- ▶ the simply-laced Dynkin diagram;
- ▶ the involution τ ;
- ▶ the fixed-point Lie algebra $\mathfrak{g}^\theta$.



Example: type A with diagram symmetry

Type AI : $\tau = \text{Id}$
(sl_n, so_n)

New ingredient

The involution must act simultaneously on the Lie algebra and on the loop parameter z .

The loop-group involution

Define an anti-involution on the loop group by

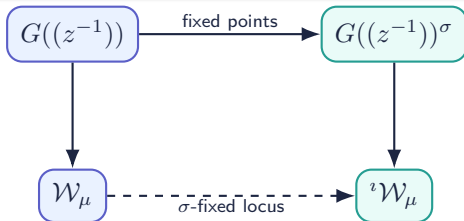
$$\sigma(g(z)) = \omega_{\tau}(g(-z))^{-1}, \quad \omega_{\tau} = \omega \circ \tau.$$

On the Lie algebra level one has the corresponding involution

$$\omega_{\tau}(xz^k) = \omega_{\tau}(x)(-z)^k.$$

Poisson compatibility

The involution σ is compatible with the Poisson structure coming from the Manin triple. Consequently, fixed point loci inherit Poisson geometry.



Even spherical coweights

The slice $\mathcal{W}_\mu$ is preserved by σ precisely under a parity condition on μ .

Condition

The coweight μ is called **even spherical** when

$$\mu = \mu_1 + \tau\mu_1.$$

This is the natural condition ensuring that σ preserves the slice indexed by μ .

Fixed-point slices

For even spherical μ and $\lambda = \tau\lambda \geq \mu$, define

$${}^i\mathcal{W}_\mu = (\mathcal{W}_\mu)^\sigma, \quad {}^i\overline{\mathcal{W}}_\mu^\lambda = (\overline{\mathcal{W}}_\mu^\lambda)^\sigma.$$

These are the affine Grassmannian fixed point slice and islice.

What changes after taking fixed points?

Ordinary world

$$Y_\mu \rightsquigarrow \mathcal{W}_\mu, \quad Y_\mu^\lambda \rightsquigarrow \overline{\mathcal{W}}_\mu^\lambda.$$

- ▶ Yangians
- ▶ affine Grassmannian slices
- ▶ type A Slodowy slices

Twisted world

$${}^iY_\mu \rightsquigarrow {}^i\mathcal{W}_\mu, \quad {}^iY_\mu^\lambda \rightsquigarrow {}^i\overline{\mathcal{W}}_\mu^\lambda.$$

- ▶ twisted Yangians
- ▶ affine Grassmannian islices
- ▶ classical B, C, D Slodowy slices

Moral

The twist turns “type A slice geometry” into symmetric-pair slice geometry.

Shifted twisted Yangians

Adding a twist to the Yangian–slice dictionary

Twisted Yangians and shifted twisted Yangians

Twisted Yangians arise as quantum analogues of the current algebra fixed by a symmetric pair:

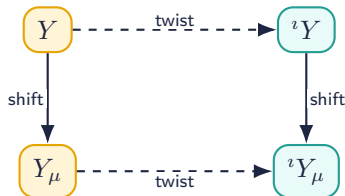
$${}^iY \rightsquigarrow U(\mathfrak{g}[z]^{\omega_\tau}).$$

Shifted version

For an even spherical coweight μ , one defines a shifted twisted Yangian

$${}^iY_\mu.$$

It is the analogue of the ordinary shifted Yangian Y_μ .



PBW and the classical shadow

The first structural result is that shifted twisted Yangians behave like honest deformations.

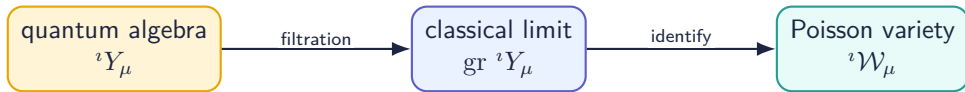
PBW theorem

The algebra ${}^iY_\mu$ has a PBW basis. In particular, it has a well-controlled associated graded algebra.

Why PBW matters

PBW is the algebraic mechanism that lets us compare:

noncommutative generators and relations $\longleftrightarrow$ coordinate functions on fixed point varieties.



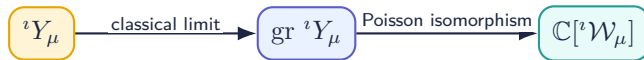
Main theorem: quantization of fixed point loci

Theorem

Let μ be even spherical. Then there is a filtration on ${}^iY_\mu$ such that

$$\mathrm{gr} \, {}^iY_\mu \cong \mathbb{C}[{}^i\mathcal{W}_\mu]$$

as graded Poisson algebras.



Interpretation

Shifted twisted Yangians are quantizations of affine Grassmannian fixed point slices.

iGKLO representations and truncations

The ordinary GKLO homomorphism realizes shifted Yangians by difference operators. The twisted story has an $\imath$ -analogue:

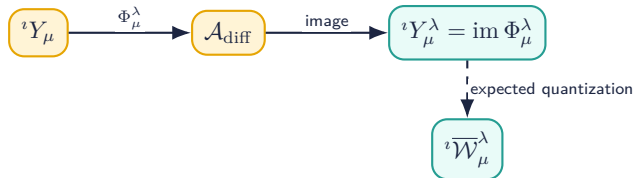
$$\Phi_\mu^\lambda : {}^\imath Y_\mu \longrightarrow \mathcal{A}_{\text{diff}}.$$

Truncated shifted twisted Yangian

Define

$${}^\imath Y_\mu^\lambda := \text{im } \Phi_\mu^\lambda.$$

This quotient is the twisted analogue of the truncated shifted Yangian.



What is proved, and what is expected?

Assume μ is even spherical and $\lambda = \tau\lambda \geq \mu$.

Proved

There is a filtration on ${}^iY_\mu$ with

$$\mathrm{gr} \, {}^iY_\mu \cong \mathbb{C}[{}^i\mathcal{W}_\mu].$$

Moreover the iGKLO image controls an open part of the islice.

Expected

The truncated shifted twisted Yangian ${}^iY_\mu^\lambda$ should quantize a top-dimensional irreducible component

$$\widehat{\mathcal{C}}_\mu^\lambda \subset {}^i\overline{\mathcal{W}}_\mu^\lambda.$$

Technical bottleneck

The conjecture reduces to comparing two natural filtrations on the same algebraic object.

Type AI and classical Slodowy slices

Adding a twist to the Yangian–slice dictionary

Type AI: the cleanest test case

Take

$$G = PGL_N, \quad \tau = \text{id}.$$

The involution on $\mathfrak{sl}_N$ has the form

$$\sigma_\varepsilon : X \mapsto -J_\varepsilon^{-1} X^t J_\varepsilon,$$

$$\mathfrak{sl}_N^\varepsilon := \mathfrak{sl}_N^{d_\varepsilon} = \begin{cases} \mathfrak{so}_N & \varepsilon = +1 \\ \mathfrak{sp}_N & \varepsilon = -1 \end{cases}$$

where J_ε is symmetric or skew-symmetric depending on $\varepsilon = \pm 1$.

$$\varepsilon = +1$$

Orthogonal fixed points: type B/D behavior.

$$\varepsilon = -1$$

Symplectic fixed points: type C behavior.

Key point

One can choose J_ε compatibly with the $\mathfrak{sl}_2$ -triple, so Slodowy slices inherit the same fixed-point involution.

preserving Slodowy slices

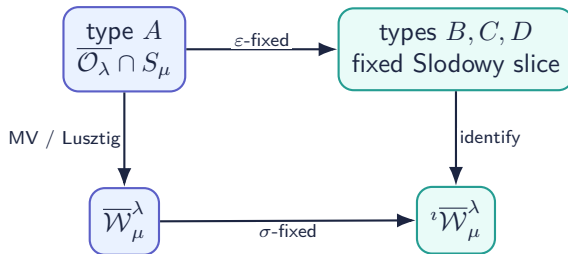
Type AI: islices as classical Slodowy slices

Let μ be viewed as a partition. Under the parity condition appropriate to ε , the fixed point Slodowy slice matches the affine Grassmannian islice.

Theorem: type AI

For suitable λ, μ ,

$$(\overline{\mathcal{O}}_\lambda \cap S_\mu)^\varepsilon \cong (\overline{\mathcal{W}}_\mu^\lambda)^\sigma = {}^i\overline{\mathcal{W}}_\mu^\lambda.$$



Parity in type AI

The partition condition on μ determines whether the fixed slice lies in the orthogonal or symplectic case.

Sign	Fixed-point type	Partition condition on μ
$\varepsilon = +1$	orthogonal	even parts occur with even multiplicity
$\varepsilon = -1$	symplectic	odd parts occur with even multiplicity

Recall type A $G = \mathrm{PGL}_N$

$$\begin{array}{ccc}
 \overline{O}^\lambda & \xrightarrow{\sim} & \overline{W}_0^\lambda \\
 \cap & \text{[Lusztig]} & \cap \\
 \mathcal{N}_{\mathrm{SL}_N} & \xrightarrow{\sim} & \overline{W}_0^{N\omega_1^\vee} \\
 \cup & & \cup \\
 S_\mu & \xrightarrow{\sim} & \overline{W}_\mu^{N\omega_1^\vee} \\
 \cup & \text{[MV]} & \cup \\
 \overline{O} \cap S_\mu & \xrightarrow{\sim} & \overline{W}_\mu^\lambda
 \end{array}$$

Theorem [LWW]

$$\begin{array}{ccc}
 \overline{O}_{\mathrm{SL}_N^\varepsilon}^{\lambda^+} & = (\overline{O}^\lambda)^{\delta_\varepsilon} \xrightarrow{\sim} (\overline{W}_0^\lambda)^{\delta_\varepsilon} =: \overline{W}_0^\lambda & \lambda^+ \trianglelefteq_{\max} \lambda \\
 \cap & & \cap \\
 \mathcal{N}_{\mathrm{SL}_N^\varepsilon} & \xrightarrow{\sim} & \overline{W}_0^{N\omega_1^\vee} \\
 \cup & & \cup \\
 S_\mu^{\mathrm{SL}_N^\varepsilon} & = (S_\mu)^{\delta_\varepsilon} \xrightarrow{\sim} (\overline{W}_\mu^{N\omega_1^\vee})^{\delta_\varepsilon} =: \overline{W}_\mu^{N\omega_1^\vee} \\
 \cup & & \cup \\
 \overline{O}_{\mathrm{SL}_N^\varepsilon}^{\lambda^+} \cap S_\mu^{\mathrm{SL}_N^\varepsilon} & = (\overline{O}^\lambda \cap S_\mu)^{\delta_\varepsilon} \xrightarrow{\sim} (\overline{W}_\mu^\lambda)^{\delta_\varepsilon} =: \overline{W}_\mu^\lambda
 \end{array}$$

$\mu \begin{cases} \text{all parts odd} & \varepsilon = +1 \\ \text{all parts even} & \varepsilon = -1 \end{cases}$

Coulomb branches and outlook

Adding a twist to the Yangian–slice dictionary

Coulomb-branch expectation

Ordinary affine Grassmannian slices are Coulomb branches of framed ADE quiver gauge theories.

Ordinary story

$$\overline{\mathcal{W}}_{\mu}^{\lambda} \cong \mathcal{M}(G_{\mathbf{V}}, \mathbf{E}_{\mathbf{V}, \mathbf{W}}).$$

$$W \rightsquigarrow \lambda = \sum w_i \omega_i$$

$$V \rightsquigarrow \lambda - \mu = \sum v_i \alpha_i$$

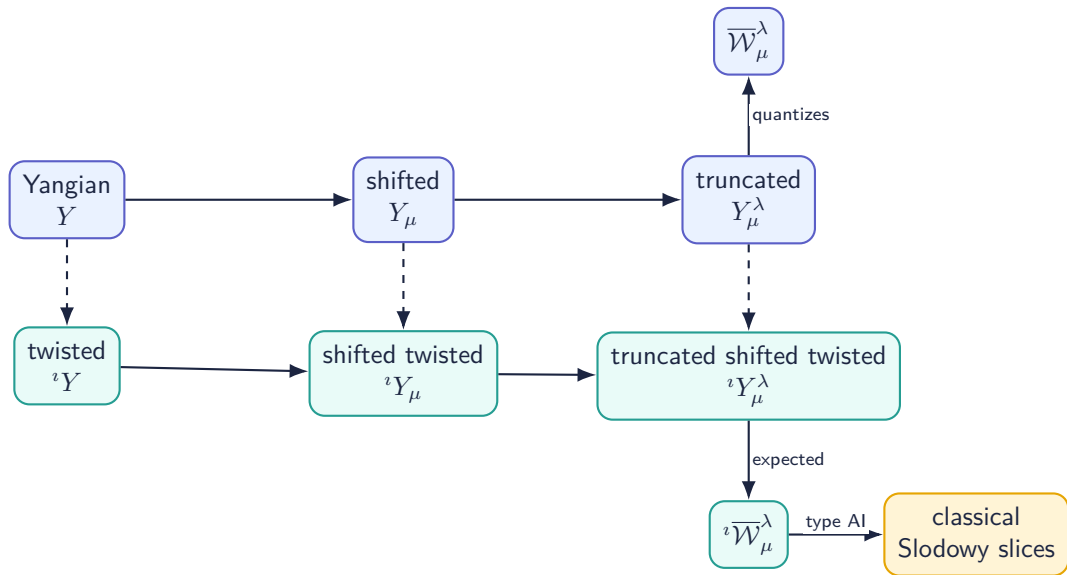
Twisted expectation

The slice should be normalized by an iCoulomb branch built from Satake framed double quivers:

$$\mathcal{M}^i(\mathbf{V}, \mathbf{W}) \longrightarrow {}^i\overline{\mathcal{W}}_{\mu}^{\lambda}.$$

For split Satake data this produces ortho-symplectic Coulomb branches; nonsplit data leads to hybrid variants.

Final picture



Takeaways

1. Affine Grassmannian slices admit natural symmetric-pair fixed loci.
2. These fixed loci are the affine Grassmannian islices.
3. Shifted twisted Yangians ${}^{\iota}Y_{\mu}$ quantize the fixed point slices ${}^{\iota}\mathcal{W}_{\mu}$.
4. Truncated shifted twisted Yangians ${}^{\iota}Y_{\mu}^{\lambda}$ are expected to quantize top-dimensional pieces of ${}^{\iota}\overline{\mathcal{W}}_{\mu}^{\lambda}$.
5. In type AI, the geometry recovers Slodowy slices of classical types B, C, D .

One-sentence summary

The paper builds the ι -analogue of the Yangian–slice dictionary.

Thank you!

Questions?

Backup slides

Adding a twist to the Yangian–slice dictionary

Backup: Manin triple and Poisson structure

The Lie algebras

$$\mathfrak{g}((z^{-1})), \quad \mathfrak{g}[z], \quad z^{-1}\mathfrak{g}[[z^{-1}]]$$

form a Manin triple with respect to

$$(x, y) = -\operatorname{Res}_{z=0}(x, y)_{\mathfrak{g}} dz.$$

Consequence

The loop groups $G((z^{-1}))$, $G[z]$, and $G_1[[z^{-1}]]$ carry compatible Poisson structures. This is the Poisson-geometric origin of the structures on $\mathcal{W}_{\mu}$ and ${}^i\mathcal{W}_{\mu}$.

Backup: coordinate functions and filtrations

On $\mathcal{W}_\mu$ one has distinguished coordinate functions

$$e_i^{(r)}, \quad f_i^{(r)}, \quad h_i^{(s)}.$$

They are the classical shadows of the Drinfeld generators

$$E_i^{(r)}, \quad F_i^{(r)}, \quad H_i^{(s)}$$

of Y_μ .

Filtered-to-Poisson principle

A filtered algebra A quantizes a Poisson algebra R when

$$\mathrm{gr} A \cong R$$

and the commutator in A induces the Poisson bracket on R .